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Dear MITRE Hiring Team,

I am writing to express my interest in MITRE's summer internship program, with particular interests in operations research and applied data science. I am a junior at Johns Hopkins University pursuing a B.S. in Applied Mathematics and Statistics (AMS), a B.A. in Economics, and an M.S.E. in AMS. I am drawn to problems that require translating complex systems into mathematical models and using data, optimization, and computation to improve decision-making—work that aligns closely with MITRE's technical focus and public-service mission.

As a Research Assistant in Dr. Domenico Giannone's lab at Johns Hopkins, I apply statistical and machine learning methods to macroeconomic forecasting using Python and PyTorch. My current research focuses on predicting the distribution of future changes in the U.S. unemployment rate using FRED-MD macroeconomic data. I developed a forecasting pipeline that processes more than 100 economic indicators and implements recursive quantile forecasting with linear and deep neural network models. Through this work, I have learned how to translate statistical concepts into computational methods, work through the challenges of real-world data, and evaluate the resulting models.

My coursework has also provided a strong foundation in optimization, probability, statistics, computational mathematics, time series analysis, and data science. I am particularly interested in optimization and operations research problems where mathematical models can improve how systems operate or limited resources are allocated. In optimization coursework, I have implemented algorithms including a simplex solver and formulated linear programs for scheduling and resource-allocation problems. These experiences directly motivate my interest in MITRE's work in applied mathematics and optimization, simulation and computational analytics, systems modeling, and augmented decision-making.

MITRE is especially compelling to me because it applies these methods to problems with meaningful public impact. I want to build a career using statistical and analytical expertise to improve decision-making and the systems people rely on. The opportunity to contribute to challenges across national security, healthcare, transportation, and citizen services while learning from MITRE's researchers would allow me to develop technically while contributing to meaningful work in the public interest.

Beyond my technical experience, competing as an NCAA swimmer at Johns Hopkins has taught me persistence, discipline, and collaboration. I would be excited to bring that mindset and my background to a MITRE team. Thank you for considering my application, and I look forward to the opportunity to discuss how my experience and interests could contribute to MITRE's work.

Sincerely,
Josiah March